

AARON CHEUNG

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Plano, TX | Open to Relocation

EDUCATION

University of Texas at Dallas - B.S. in Computer Science
Collin College - A.S., Magna Cum Laude

Jan 2023 - Dec 2025
Aug 2022 - May 2025

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, C++, HTML, CSS
Frameworks & Libraries: React, Node.js, Express, Tailwind CSS
Tools & Platforms: Git, GitHub, Supabase, Clerk, MySQL, Selenium, Jira
Design: UI/UX Design, Figma, Krita

EXPERIENCE

The Review Mediator

Fall 2025

Capstone Full-Stack Developer - Plano, TX

- **Built** a full-stack review-management dashboard with a team of 6, rotating through backend, database, and frontend roles before closing out the platform's UI directly with the product owner.
- **Engineered** a responsive front end in **React**, **Tailwind CSS**, and **Flowbite**, wiring a full-stack contact form through **Express** middleware into a **Supabase** backend.
- **Secured** the platform with dual-layer access control - validating **JWTs** in Express middleware and enforcing **Supabase row-level security** policies alongside **Clerk**-based role routing - so each business could only ever access its own data.

PROJECTS

All projects live at aaroncheung.me

Personal Portfolio - aaroncheung.me

HTML, CSS, JavaScript

- **Engineered** a terminal/TUI-style portfolio site from scratch with zero frameworks, libraries, or build step, driving all content through a structured **JSON** data layer.
- **Built** a Ctrl+K fuzzy-search command palette that indexes every page on the site as a live navigator, paired with full keyboard navigation and a persisted theming system spanning 4 color palettes (default, ocean, forest, ember) plus a CRT display mode.
- **Audited** the site for accessibility and performance, fixing real issues including a forced-layout-recalculation animation bug, a WCAG contrast failure, and a zero-size touch-target bug.

Bitboard Game AI - Jumpy

C++, JavaScript

- **Solved** a 1x8 strategy board game completely via retrograde analysis / backward induction over the full reachable state space, guaranteeing optimal play from any position.
- **Packed** the entire game state into a single 16-bit integer using **bitboards**, enabling fast state transitions and lookups across the solved-game table.
- **Layered** an alpha-beta-pruned iterative deepening search (depth 8-10) beneath the solver to power a live "watch it think" visualization and handle fallback logic for undetermined draw-by-repetition positions.

Lofi Generator

JavaScript, Web Audio API

- **Built** a procedural lofi beat generator, driving procedurally-generated chords, melody, and ambience through a functional-harmony engine over the raw **Web Audio API**, layered with sampled piano and drum hits.
- **Modeled** harmonic movement across a 7-chord diatonic transition table, with the melody engine walking a 15-note scale range in chord-tone-aware steps across 4 rhythmic subdivisions (half/quarter/eighth/sixteenth).
- **Mixed** three independently-layered ambience beds - precipitation, atmosphere, and vinyl crackle - each synthesized live and decoupled from the music engine.

Relational Database System

MySQL, PHP

- **Designed** a normalized relational schema spanning 18+ tables modeling product manufacturing, part procurement, and workforce data.
- **Shipped** a PHP API exposing 19 endpoints - 15 required queries plus insert/delete/lookup actions - running joins, aggregations, and live writes directly from the browser.
- **Implemented** relational-division queries (e.g. salesmen who sold every product type priced above \$200) via nested NOT EXISTS / NOT IN subqueries.